THE BLOWUP OVER AN AFFINE OPEN IS THE PROJ OF THE REES ALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The card was recovered after a quality read: the prefilter dropped the candidate for the phrase 'The first statement is clear from the construction of the relative Proj via glueing', which names the construction that gives it, while the affine-cover statement is argued in place.

Editorial note. This lemma is the affine input used by several cards of this pool (for instance L44203, L44207, L44212, L44218), where it enters as a navigation pointer.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Construction of the relative Proj by glueing. Constructions, Section constructions-section-relative-proj-via-glueing, used at proof lines 8336-8338. Gives the identification of b^{-1}(U) with the Proj of the Rees algebra.
- Definition of the affine blowup algebra. Algebra, Definition algebra-definition-blow-up, used at proof lines 8342-8345. Identifies D_+(a^{(1)}) with Spec(A[I/a]).

Source reference. Original definition 052Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6578–6578. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 01NM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2163–2163. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a scheme. Let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals. Let $U = \mathrm{Spec}(A)$ be an affine open subscheme of X and let $I \subset A$ be the ideal corresponding to $\mathcal{I}|_U$. If $b : X' \to X$ is the blowup of X in $\mathcal{I}$, then there is a canonical isomorphism*

$$b^{-1}(U) = \mathit{Proj}(\bigoplus\nolimits_{d \geq 0} I^d)$$

of $b^{-1}(U)$ with the homogeneous spectrum of the Rees algebra of I in A. Moreover, $b^{-1}(U)$ has an affine open covering by spectra of the affine blowup algebras $A[\frac{I}{a}]$.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The first statement is clear from the construction of the relative Proj via glueing, see Constructions, Section 01NM. For $a \in I$ denote $a^{(1)}$ the element a seen as an element of degree 1 in the Rees algebra $\bigoplus_{n\geq 0} I^n$. Since these elements generate the Rees algebra over A we see that $\mathrm{Proj}(\bigoplus_{d\geq 0} I^d)$ is covered by the affine opens $D_+(a^{(1)})$. The affine scheme $D_+(a^{(1)})$ is the spectrum of the affine blowup algebra $A' = A[\frac{I}{a}]$, see Algebra, Definition 052Q. This finishes the proof. $\square$